

TREE DETECTION AUTOMATION LEVERAGING SATELLITE IMAGERY

A Project Report

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CERTIFICATE

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"Gratitude to all who contributed; this project report is a testament to our collective effort and commitment to excellence."

-Mark Twain

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Abstract

The "Tree Detection Automation Leveraging Satellite Imagery" project aims to revolutionize the process of tree enumeration within forested areas designated for diversion, leveraging advanced image analytics technologies. Forest land diversion plays a crucial role in various developmental projects, yet ensuring the sustainable management of forest resources necessitates accurate assessment of tree populations affected by such diversions. Traditional methods of tree enumeration are often time-consuming, subjective, and prone to inaccuracies. In response, this project proposes the adoption of cutting-edge image analytics techniques, including remote sensing, machine learning, and computer vision, to automate and enhance the tree counting process. The primary objectives include developing robust algorithms for tree detection, delineation, and enumeration; improving accuracy and efficiency; and facilitating informed decision-making regarding land use planning and environmental management. Through interdisciplinary collaboration and rigorous validation, the project seeks to deliver a scalable and user-friendly solution that empowers stakeholders with actionable insights while promoting transparency and accountability in forest land diversion activities. This report provides an overview of the project's methodologies, challenges, outcomes, and recommendations for future research and application.

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Chapter 1

Introduction

1.1 Project introduction

The project titled "Tree Detection Automation Leveraging Satellite Imagery" represents a significant endeavor aimed at revolutionizing the process of tree enumeration within forested areas designated for diversion. Forest land diversion plays a crucial role in various developmental projects such as infrastructure development, urban expansion, and industrialization. However, ensuring the sustainable management of forest resources and minimizing environmental impacts necessitates accurate and comprehensive assessment of tree populations affected by such diversions. Traditional methods of tree enumeration, relying heavily on manual surveys and ground-based assessments, often suffer from limitations such as time constraints, subjective biases, and logistical challenges. Moreover, the sheer scale and complexity of forested landscapes make it increasingly challenging to obtain reliable data through conventional means.

In response to these challenges, the project proposes the adoption of advanced image analytics techniques to automate and enhance the process of tree enumeration. By leveraging cutting-edge technologies such as remote sensing, machine learning, and computer vision, the project aims to facilitate rapid, accurate, and cost-effective assessment of tree densities and distribution patterns across large swathes of forested land.

1.2 Scope of the Project

The scope of the "Tree Detection Automation Leveraging Satellite Imagery" project encompasses several key aspects aimed at achieving its objectives effectively and efficiently. The project's scope includes, but is not limited to, the following components:

1. Data Acquisition and Preprocessing
2. Algorithm Development

3. Model Training and Validation
4. Integration with Geospatial Technologies
5. Performance Evaluation and Optimization
6. User Interface Development
7. Field Validation and Ground Truthing
8. Documentation and Knowledge Transfer

1.3 Aim of the Project

The primary aim of the "Tree Detection Automation Leveraging Satellite Imagery" project is to develop and implement advanced image analytics techniques to streamline and enhance the process of tree enumeration within forested areas designated for diversion. This overarching goal encompasses several specific objectives:

1. Automation of Tree Enumeration
2. Enhanced Accuracy and Reliability
3. Efficiency Improvement
4. Support for Decision-making
5. Environmental Conservation
6. Technological Innovation

Chapter 2

Literature Survey

2.1 Tree counting with high spatial-resolution satellite imagery based on deep

Author: Ling Yao Tang Liu Jun Qin Ning Lu

Abstract:

Forest inventory at single-tree level is of great importance to modern forest management. The inventory contains two critical parameters about trees, including their numbers and spatial locations. Traditional methods to catalogue single trees are laborious, while deep neural networks enable to discover the multi-scale features hidden in images and thus make it possible to count trees with remote sensing imagery. In this study, four different tree counting networks, which were constructed by remodeling four different classical deep convolutional neural networks, were evaluated to determine their abilities to grasp the relationship between remote sensing images and tree locations for automatic tree counting end-to-end. To this end, a tree counting dataset was constructed with remote sensing images of 0.8-m spatial resolution in distinct regions.

2.2 A novel image processing technique for counting the number of trees in a satellite image

Author: D. Girish Kumar M. Padmaja

Abstract:

This Paper Presents, an Image Processing Method for counting the number of trees in a High Spatial Resolution Satellite Image using Texture Approach. A hierarchical strategy is followed for detect, delineate and counting the trees in an image. Texture Approach Algorithm converts the RS Image in to HSI Image for more qualitatively analyze the image data. Image delineation is done

approximately for using existing methods and also not gives the accurate tree count. This Algorithm gives accurate image delineation and also gives accurate tree count with the usage of Structured Element in Image. The fundamental process of an algorithm, simulation results are explained with image examples and limitations are discussed

2.3 Automatic extraction of trees from aerial images and surface models

Author: Bernd-M. Straub

Abstract:

In this paper we present an approach for the automatic extraction of trees and the boundaries of the tree's crowns. The approach is based on a multi-scale representation of an orthoimage and a surface model in Linear Scale Space. The idea of the approach is that the coarse structure of the crown can be approximated with the help of a sphere or an ellipsoid. This assumption is true, if the fine structure of the crown is ignored and the coarse structure is revealed in an appropriate level of the multi-scale representation of the surface model. But this scale level is unknown, because it is correlated with the unknown diameter of the crown. The proposed solution of this chicken-and-egg problem is to investigate a wide range of scale levels, and to select the best hypothesis for a crown from all these scale levels. The segmentation of the surface model is performed using the watershed transformation. The boundary of every crown is measured with Active Contours (Snakes). The approach was tested with surface models of different resolutions (0.25 m and 1 m) and different sensors, laser scanner and image matching. An overview of the approach is given in the paper and important points are discussed.

2.4 Survey on Image Enhancement Techniques

Author:P.Suganya ,S.Gayathri,N.Mohanapriya

Abstract:

Enhancement is one of the challenging factors in image processing. The objective of enhancement is to improve the structural appearance of an image without any degradation in the input image. The enhancement techniques make the identification of key features easier by removing noise and other artifacts in an image. This paper analyzes the performance of various enhancement techniques based on noise ratio, time delay and quality. It also suggest suitable algorithm for remote sensing images based on the analysis.

2.5 Automated tree crown detection and delineation in high-resolution digital camera imagery of coniferous forest regeneration

Author: Douglas J. King, D.A. Pouliot, F. Wayne Bell

Abstract:

Ensuring successful forest regeneration requires an effective monitoring program to collect information regarding the status of young crop trees and nearby competing vegetation. Current field-based assessment methodology provides the needed information, but is costly, and therefore assessment frequency is low. This often allows undesirable forest structures to develop that do not coincide with management objectives. Remote sensing techniques provide a potentially low-cost alternative to field-based assessment, but require the development of methods to easily and accurately extract the required information. Automated tree detection and delineation algorithms may be an effective means to accomplish this task. In this study, a tree detection–delineation algorithm designed specifically for high-resolution digital imagery of 6-year-old trees is presented and rigorously evaluated. The algorithm is based on the analysis of local transects extending outward from a potential tree apex. The crown boundary is estimated using the point of maximum rate of change in the transect data and a rule base is applied to ensure that the point is contextually suitable. This transect approach is implemented in both the tree-detection and crown-delineation phases. The tree-detection algorithm refines the results of an initial local maximum filter by providing an outline for each detected tree and retaining only one local maximum value within this outline.

2.6 Self-supervised approach for Urban Tree Recognition on Aerial Images

Author: Lakshmi Babu Saheer and Mohamed Shahawy

Abstract:

Artificial Intelligence is helping us combat climate change! This study explores how deep learning can spot vegetation in aerial photos. It's the initial phase of a larger scheme that includes an eco-monitoring system. This system would monitor traffic, weather, air quality, and even the number of trees in towns all across the world. There's a catch, though: getting the tree-finding model to function in recently developed cities with scant labeled data. The solution suggested by the researchers is self-supervised learning. Their technique demonstrated an exceptional mean average precision of 0.89 when tested on Google Earth photos from Cambridge.

2.7 A benchmark dataset for canopy crown detection and delineation in co-registered airborne RGB, LiDAR and hyperspectral imagery from the National Ecological Observation Network

Author: Ben G. Weinstein , Sarah J. Graves, Sergio Marconi, Aditya Singh

Abstract:

New technology using airplanes with fancy sensors can now map entire forests in detail. However, pinpointing each tree's exact location can be challenging. Numerous computer programs, or algorithms, have been developed to address this issue; but, since each person utilizes a distinct set of data, it is impossible to determine which approach performs best.

Scientists created a massive database of forest data from aircraft to address this. This collection contains laser measurements (LiDAR), images (RGB), and even hyperspectral light readings. Still, that's not all! They also contained actual data that was gathered from thousands of reference points and real trees on the ground. They included a unique computer application to assist in comparing all these various tree-finding systems, just to make things even easier.

2.8 A remote sensing derived data set of 100 million individual tree crowns for the National Ecological Observatory Network

Author: Ben G. Weinstein , Sarah J. Graves, Aditya Singh, Stephanie A Bohlman

Abstract:

Many ecological and economic advantages of forests exist, but comprehending them necessitates knowing specific tree data. Logistically and financially, gathering this data has always been difficult. Though there are technical obstacles in turning the raw sensor data into useful information, recent advances in remote sensing provide previously unheard-of insights into individual trees.

In order to fill this gap, this study created an open-source dataset by utilizing deep learning techniques. With data collected by the National Ecological Observatory Network's Airborne Observation Platform, the collection includes crown estimates for an astounding 100 million trees spread over 37 locations. The height, crown area, and exact location of every tree crown are all precisely recorded by a bounding box.

2.9 Multisensor Data Fusion for Improved Segmentation of Individual Tree Crowns in Dense Tropical Forests

Author: Méline Aubry-Kientz, Anthony Laybros, Ben Weinstein, James G. C. Ball, Toby Jackson

Abstract:

It is still difficult to precisely segment individual tree crowns in tropical forests using remote sensing, especially when the canopies are dense, diversified, and multi-layered. The goal of this project is to determine whether segmentation accuracy in French Guiana may be increased by combining multispectral and hyperspectral data with Airborne Laser Scanning (ALS) data.

The suggested method achieves an amazing 83

Additionally, the study shows that this approach may be used to map tree mortality through follow-up surveys. Estimating the number of dead trees in such zones was successful when the overlap between previously detected crowns and areas showing clusters of height reduction was examined.

2.10 Semi-Supervised Deep Learning Neural Networks Individual Tree-Crown Detection in RGB Imagery Using

Author: Stephanie A Bohlman, Ben G. Weinstein, Ethan White , Alina Zare , Sergio Marconi

Abstract:

For extensive surveys of forestry and biodiversity, remote sensing provides a ground-breaking instrument. But the amount of data is often too great for us to manually classify individual trees in high-resolution photography. By putting forth a semi-supervised deep learning strategy for tree crown identification in RGB (color) imaging, this research addresses this problem.

Determining the exact borders of individual tree crowns via remote sensing has long been a challenge because current algorithms produce inconsistent results. To tackle this problem, this work makes use of deep learning. The suggested technique makes use of already-existing, unlabeled Light Detection and Ranging (LiDAR) data to produce a preliminary, unsupervised mapping of possible tree sites. This data, despite being fundamentally "noisy," provides a useful foundation for the neural network, enabling it to pick up fundamental characteristics linked to trees.

This work demonstrates how semi-supervised deep learning can help overcome the drawbacks of labeled training data in the context of distant sensing. Large-scale ecological surveys can be conducted more effectively by deliberately combining freely available unlabeled data with a limited

amount of high-quality labeled data to train robust AI models.

2.11 UAV-Based Computer Vision System for Orchard Apple Tree Detection and Health Assessment

Author: Hela Jemaa, Wassim Bouachir , Brigitte Leblon, Armand LaRocque

Abstract:

Timely and accurate inventories of orchards are essential to efficient management techniques. Traditional tree surveying techniques are time-consuming and error-prone, and they involve counting, finding, and evaluating the health of trees. For automated tree detection and health analysis, this study suggests a unique framework that makes use of unmanned aerial vehicles (UAVs) and computer vision algorithms.

Two crucial phases are utilized by the suggested method:

Tree Detection: With RGB imagery from the UAV's capture, a strong tree detection model is created. "Hard negative mining" is an approach this model uses to increase its accuracy. Plant

Index Derivation: Multi-band imaging is used for health classification. The health status of trees is then classified using these indices and additional pertinent data.

The study also emphasizes the possibility of wider applicability in different plantation environments. Precision agriculture will advance thanks to the framework, which provides a potential path for automated plant detection and health evaluation using UAV technology.

2.12 Accuracy Assessment and Impact Factor Analysis of GEDI Leaf Area Index Product in Temperate Forest

Author: Cangjiao Wang, Duo Jia, Shaogang Lei, Izaya Numata, Luo Tian

Abstract:

Because forests are so important to controlling the planet's temperature, scientists need a means to gauge their health. One instrument is the Leaf Area Index (LAI), which provides information on the amount of leaf surface area in a forest. The goal of the GEDI satellite mission is to map LAI worldwide, but how reliable are the readings, particularly in temperate forests with a variety of tree species?

The LAI estimates (GEDI LAIe) from GEDI and comprehensive measurements from ground-based lasers in temperate woods across the United States were compared in this work. With an average bias of -0.56 square meters per square meter, they discovered that GEDI LAIe tends to

underestimate the true leaf area. The accuracy is broken down as follows:

0.70 square meters per square meter is the mean absolute error (MAE).

0.89 square meters per square meter is the root mean square error (RMSE).

It's interesting to note that underestimating was higher in mixed and broadleaf woods than in coniferous ones. Subsequently, researchers looked at what influences GEDI LAIe accuracy. The primary cause was found to be the discrepancy in the way ground-based lasers and GEDI measure "gaps" between trees. Lesser degrees of influence was also given by geography, soil properties, and the structure of the forest canopy.

2.13 Very high resolution canopy height maps from RGB imagery using self-supervised vision transformer and convolutional decoder trained on aerial lidar

Author: Jamie Tolan, Hung-I Yang, Benjamin Nosarzewski, Guillaume Couairon, Huy V. Vo, John Brandt

Abstract:

Mapping the structure of vegetation is essential for comprehending the global carbon cycle and tracking natural methods of mitigating and adapting to climate change. It is possible to see deforestation or deterioration of existing forests, natural forest regeneration, and the use of sustainable farming methods such as agroforestry through repeated observations of these data. Because forest structures can be substantially spatially variable, particularly in agroforestry systems, assessments of tree canopy height and crown projected area at a high spatial resolution are therefore critically relevant for monitoring carbon fluxes and evaluating tree-based land uses. Information at the tree level can be extracted from very high resolution satellite photography (less than one meter, or one m Ground Sample Distance), which also enables very large-scale monitoring.

This study offers the first high-resolution canopy height map created for several sub-national jurisdictions at the same time. We generate very high resolution canopy height maps for the states of California and São Paulo, which is a notable advance over the previous global maps of canopy height based on Sentinel and GEDI, which had a resolution of ten meters (10 m). A dense prediction decoder is trained against aerial lidar data to create the maps, which are produced by extracting features from a self-supervised model trained on Maxar imagery from 2017 to 2020. Using a neural network trained using GEDI data, we additionally include a post-processing step. We assess the suggested maps by contrasting them with other remotely sensed maps, field-collected

data, and set-aside validation lidar data.

2.14 Crown-CAM: Reliable Visual Explanations for Tree Crown Detection in Aerial Images

Author : Mojtaba Marvasti, Devin Goodsman

Abstract:

The ability to visually explain "black-box" models has made it possible for artificial intelligence (AI) specialists and researchers to significantly increase the localization potential of these techniques. While most current visual explanation approaches have been applied to single item classification tasks, their applicability to detection tasks, where issues may extend beyond basic coarse area-based discriminating, has not been thoroughly investigated. This is especially crucial when a detector is supposed to be facing many items of varied sizes from different angles, or if the things of interest aren't there. In this study, we offer CrownCAM, a visual explanation tool that can produce trustworthy answers to the dynamic and difficult problem of tree crown detection in aerial photos.

It effectively provides non-contextual backdrop suppression and fine-grain localization of tree crowns for landscapes lacking tree crowns or settings with extremely dense forest trees in the face of possible distractions. Two Intersection over Union (IoU) based metrics are also presented, which can be used to measure the precision and imprecision of generated visual explanations for areas in the picture that have or do not have tree crowns. Evaluations based on empirical data show that the proposed Crown-CAM improves the accuracy (and reduces the inaccuracy) of visual explanations on the difficult NEON tree crown dataset by an average IoU margin of 8.7, 5.3, and 21.7 (and 3.3, 9.8, and 16.5), respectively.

2.15 Implementing Deep Learning algorithms for urban tree detection and geolocation with high-resolution aerial, satellite, and ground-level images

Author: Luisa Fernanda Velasquez , Maddi Etxegarai

Abstract:

Due to their ability to provide ecosystem services that both mitigate climate change and enhance the well-being of city people, urban forests are becoming more and more significant for human well-being. But because standard forest surveys in urban settings are expensive and difficult to

carry out, there is a lack of information about most urban forests in the globe, despite their significance. Artificial intelligence-based innovations have the potential to be a wise and effective replacement for expensive traditional inventory. The method we describe in this study uses a combination of aerial/satellite and ground-level data to recognize, count, and geoposition trees. It depends on deep learning techniques.

We experimented with multiple convolutional networks, varying the query distance between ground-level photos, the detection radius, and the resolution of satellite and aerial images, as well as investigating alternative combinations of hyperparameters. With a positional accuracy of 60 centimeters to the canopy's center, our technology can identify and precisely locate 79 percent of urban street trees. This method also enables us to ascertain the availability of images of urban trees, giving the Google Street View image from which each tree may be seen. Our study highlights the potential of artificial intelligence to completely transform the way we inventory and manage urban forests by offering a scalable and replicable response to the lack of data and knowledge on urban trees available globally.

Chapter 3

Analysis / Software Requirements Specification (SRS)

3.1 Introduction

The purpose of this document is to specify the software requirements for an image analytics system designed for tree enumeration in the context of forest land diversion. The system aims to accurately identify and enumerate trees within designated forest areas to facilitate land diversion processes while ensuring compliance with environmental regulations and conservation efforts.

3.1.1 Document Conventions

1. Heading Styles: Main headings in bold, numbered format. Subheadings in bold.
2. Font Styles: Important terms italicized. Code snippets in monospaced font.
3. Acronyms: Spelled out on first use, followed by acronym in parentheses.
4. Lists: Bullet points for unordered lists. Numbered lists for ordered steps.
5. Requirements Format: Functional numbered, non-functional numbered with decimals.
6. Annotations: Notes, cautions marked appropriately.
7. References: Consistent citation format.
8. Cross-References: Hyperlinked or referenced by section number.
9. Diagrams: Labeled with descriptive captions.
10. Version Control: Document revisions documented with version numbers, dates, and brief descriptions

3.1.2 Intended Audience and Reading Suggestions

This document is intended for environmental stakeholders, forestry experts, land management agencies, and software developers involved in forest land diversion projects. It is recommended for those seeking a comprehensive understanding of the software requirements for implementing image analytics systems for tree enumeration in the context of forest land diversion. Additionally, individuals interested in environmental conservation, forestry management, and technological solutions for sustainable land use may find the document informative. Reading suggestions include focusing on relevant sections based on roles: stakeholders can review executive summaries, while developers may delve into technical requirements.

3.1.3 Product Scope

The project encompasses the development of an image analytics software system tailored for tree enumeration within forested regions targeted for land diversion. It includes modules for image acquisition, preprocessing, tree detection, species classification (optional), quantitative analysis, and a user interface. The system aims to accurately identify and enumerate trees, potentially categorizing them by species, to facilitate decision-making processes related to forest land diversion while ensuring compliance with environmental regulations and conservation efforts. It prioritizes efficiency, accuracy, and user-friendliness to meet the needs of environmental stakeholders, regulatory bodies, and land management agencies involved in forest land diversion projects.

3.2 Overall Description

3.2.1 Product Perspective

The application of image analytics for tree enumeration represents a pivotal advancement in the field of forestry and land management. By leveraging advanced image processing and analysis techniques, this technology provides a comprehensive and accurate assessment of the number, distribution, and characteristics of trees within forested areas targeted for land diversion projects. It encompasses the development of a robust and user-friendly software solution that integrates seamlessly with existing tools and systems, ensuring scalability, reliability, and compliance with relevant laws and regulations. The product aims to streamline the land diversion process, providing decision-makers with actionable insights to make informed choices while ensuring environmental sustainability and conservation. With its ability to process large volumes of data and deliver precise results, this technology represents a significant step forward in the responsible management of

forest resources.

3.2.2 Product Functions

Product functions of the application of image analytics for tree enumeration:

1. Image Processing and Analysis: The product processes high-resolution aerial or satellite images to detect, classify, and analyze trees within forested areas targeted for land diversion projects
2. Tree Detection and Classification: Utilizing advanced algorithms, the product accurately identifies and classifies different types of trees based on their visual characteristics, such as size, shape, and color.
3. Enumeration and Distribution Mapping: The product counts and maps the distribution of trees within the specified areas, providing valuable data on tree density and spatial distribution.
4. Characteristics Assessment: It assesses various characteristics of trees, including size, age, health, and species compom.

3.2.3 User Classes and Characteristics

User Classes: Forestry Professionals: This class includes foresters, arborists, and other professionals involved in forest management. They use the application to assess forest resources, plan land diversion projects, and monitor forest health. Land Managers: This class includes individuals responsible for managing land diversion projects, such as government agencies, NGOs, and private landowners. They use the application to identify suitable areas for diversion, assess the impact on forest resources, and ensure compliance with regulations.

Environmental Scientists: This class includes scientists and researchers studying forest ecosystems. They use the application to collect data on tree enumeration, analyze forest dynamics, and assess the impact of land diversion on biodiversity. GIS Analysts: This class includes professionals specializing in geographic information systems (GIS). They use the application to integrate tree enumeration data with other spatial data, such as land use, topography, and climate, to support land diversion planning and analysis.

Policy Makers: This class includes government officials and policymakers involved in forest management and land diversion. They use the application to make informed decisions on land diversion projects, based on accurate and reliable tree enumeration data.

Characteristics:

Accuracy: The application must provide accurate tree enumeration data to support informed decision-making and ensure the success of land diversion projects.

Reliability: The application must be reliable, with a low error rate and a high level of confidence in the results, to ensure the integrity of the data and the success of land diversion projects.

Scalability: The application must be scalable to accommodate large volumes of data and diverse project sizes and complexities, ensuring its usefulness for a wide range of users and applications.

Usability: The application must be user-friendly, with an intuitive interface and easy-to-use tools, to ensure that users can easily upload and process images, view results, and generate reports.

Integration: The application must integrate with other systems and tools used in land diversion projects, such as GIS software, to ensure seamless data exchange and collaboration among stakeholders.

3.2.4 Operating Environment

The operating environment for the application of image analytics for tree enumeration for the diversion of forest land is complex and requires careful consideration of hardware, software, data, network connectivity, security, regulatory compliance, user training and support, environmental conditions, collaboration and communication, and ethical considerations.

3.2.5 Design and Implementation Constraints

The operating environment for the application of image analytics for tree enumeration for the diversion of forest land involves several key components:

Hardware: The application requires hardware capable of processing large volumes of image data, such as high-performance computers or servers with sufficient processing power, memory, and storage capacity. It may also require specialized hardware, such as high-resolution cameras or drones for capturing aerial imagery.

Software: The application requires software for image processing and analysis, such as image processing libraries, machine learning frameworks, and GIS software. It may also require specialized software for tree detection, classification, and enumeration.

Data: The application requires high-resolution aerial or satellite imagery of the forested areas targeted for land diversion projects. It may also require ground-truth data, such as field surveys or remote sensing data, for validation and calibration.

Network Connectivity: The application may require network connectivity for accessing and sharing data, such as cloud-based storage or web-based applications. It may also require internet connectivity for downloading updates or accessing online resources.

Security: The application requires security measures to protect sensitive data, such as encryption of

data in transit and at rest, access controls to restrict access to authorized users, and regular security audits and updates.

3.2.6 User Documentation

User documentation for the application of image analytics for tree enumeration for the diversion of forest land would typically include the following components:

User Manual: A comprehensive guide that explains how to use the application, including step-by-step instructions for uploading images, adjusting analysis parameters, viewing results, and generating reports. The user manual should also include troubleshooting tips and frequently asked questions.

Technical Documentation: Detailed technical documentation that provides information on the algorithms and techniques used in the application, as well as the hardware and software requirements. This documentation should also include information on data formats, data sources, and data processing methods.

Training Materials: Training materials that provide hands-on exercises and examples to help users learn how to use the application effectively. This may include video tutorials, online courses, and in-person training sessions.

Support Resources: Support resources that provide users with access to technical support, such as a help desk, online forums, and a knowledge base. This should also include information on how to report bugs and request new features.

Release Notes: Release notes that provide information on new features, bug fixes, and other changes in each version of the application. This should also include information on how to update the application and access the latest version.

3.2.7 Assumptions and Dependencies

Assumptions:

The availability of high-resolution aerial or satellite imagery of the forested areas targeted for land diversion projects.

The accuracy and reliability of the analysis are dependent on the quality of the imagery and the presence of shadows, occlusions, and other visual artifacts.

The application of image analytics for tree enumeration is conducted in an ethical and environmentally responsible manner, such as minimizing the impact on wildlife and ecosystems and respecting the rights of indigenous communities.

The application complies with relevant laws and regulations governing the use of image analytics for tree enumeration and land diversion projects. The application is scalable to accommodate large volumes of data and diverse project sizes and complexities.

Dependencies:

The application depends on the availability and quality of high-resolution aerial or satellite imagery of the forested areas targeted for land diversion projects.

The accuracy and reliability of the analysis depend on the quality of the imagery and the presence of shadows, occlusions, and other visual artifacts.

The application depends on the availability of ground-truth data, such as field surveys or remote sensing data, for validation and calibration.

The application depends on network connectivity for accessing and sharing data, such as cloudbased storage or web-based applications.

The application depends on the availability of hardware capable of processing large volumes of image data, such as high-performance computers or servers with sufficient processing power, memory, and storage capacity.

3.3 External Interface Requirements

3.3.1 User Interfaces

Graphical user interface :- The Graphical User Interface (GUI) for the application of image analytics for tree enumeration in forest land diversion facilitates user interaction with the software. It offers an intuitive platform for users to upload images, configure analysis parameters, visualize results, and export reports. The GUI presents a user-friendly environment with graphical elements such as buttons, menus, and visualizations, enhancing usability. Users can navigate through various functionalities seamlessly, aided by informative feedback and progress indicators during processing tasks. The GUI ensures accessibility across different platforms (web or desktop) and prioritizes simplicity to accommodate users with varying levels of technical expertise.

Command-line interface :- The Command-Line Interface (CLI) for the application of image analytics for tree enumeration in forest land diversion provides an alternative method for interacting with the software through text-based commands. Users can execute specific commands to perform various tasks such as uploading images, configuring analysis parameters, initiating processing tasks, and exporting reports. The CLI offers flexibility and automation capabilities, allowing users to script repetitive tasks or integrate the software into existing workflows. While lacking graphical

elements, the CLI provides direct access to functionality and may appeal to users comfortable with commandbased interactions or seeking efficient, scriptable solutions.

3.3.2 Hardware Interfaces

Aerial images :- For the Application of Image Analytics for Tree Enumeration in Forest Land Diversion project, aerial images serve as a crucial hardware interface. These images are acquired through drones, satellites, or other remote sensing devices. High-resolution aerial imagery provides essential data for the image analytics system to accurately detect and enumerate trees within forested areas targeted for land diversion. The hardware interfaces ensure the availability of spatially accurate and detailed images, enabling the software to analyze tree distribution, density, and species composition effectively, thus supporting informed decision-making processes related to forest land diversion initiatives.

Geolocation data:- Geolocation data plays a vital role in the Application of Image Analytics for Tree Enumeration in Forest Land Diversion. It provides precise spatial information about the location of forested areas undergoing land diversion. By integrating geolocation data with image analytics, the system can accurately identify and enumerate trees within specific geographic coordinates. This data ensures that tree enumeration results are georeferenced, facilitating spatial analysis and decision-making processes related to forest management and land diversion projects. Geolocation data enhances the system's effectiveness in mapping tree distribution patterns, assessing habitat fragmentation, and monitoring environmental changes within targeted forest areas.

3.3.3 Software Interfaces

Integration with existing forest management systems:- In the Application of Image Analytics for Tree Enumeration for Diversion of Forest Land, seamless integration with existing forest management systems is essential. This integration ensures interoperability between the image analytics software and established forestry databases or management platforms. By interfacing with existing systems, the image analytics software can access relevant spatial and attribute data, such as forest inventory records, land cover classifications, and conservation areas. This enables comprehensive analysis and decision-making, incorporating tree enumeration results into broader forest management strategies. Integration streamlines workflows, improves data consistency, and enhances the overall efficiency of forest land diversion initiatives.

3.3.4 Communications Interfaces

File transfer protocol (FTP):- In the Application of Image Analytics for Tree Enumeration for Diversion of Forest Land, File Transfer Protocol (FTP) serves as a crucial communications interface for exchanging data between the image analytics system and external sources. FTP facilitates the secure and efficient transfer of large volumes of image data, processed results, and configuration files. Users can upload aerial or ground-level images to the system's FTP server for analysis, while the system can export processed data, reports, and visualizations back to designated FTP locations. This interface ensures seamless data exchange, enabling collaboration, data sharing, and integration with other systems involved in forest land diversion projects.

Application programming interface (API) for integration with existing systems:- In the Application of Image Analytics for Tree Enumeration for Diversion of Forest Land, an Application Programming Interface (API) facilitates seamless integration with existing forest management systems. The API allows for programmatically accessing and manipulating functionalities of the image analytics software, enabling interoperability with external systems. Through the API, forest management systems can retrieve tree enumeration data, upload images for analysis, configure analysis parameters, and retrieve processed results. This integration streamlines workflows, enhances data exchange between systems, and enables holistic forest management approaches by incorporating tree enumeration capabilities into existing platforms.

3.4 System Features

The application of image analytics for tree enumeration in the context of diverting forest land involves several key system features to ensure accurate and efficient analysis:

3.4.1 System Feature 1

Feature Name: Image Processing for Tree Enumeration

Description and Priority Description: This feature involves utilizing image analytics for tree enumeration to support the diversion of forest land. The system will analyze images to identify and count trees within specified areas.

Priority: High

Priority Components:

Benefit: 8

Penalty: 2

Cost: 7

Risk: 5

Stimulus/Response Sequences Stimulus: User uploads satellite images of the forest land. Response: System processes the images and identifies tree locations.

Stimulus: User defines the boundaries of the diversion area. Response: System focuses analysis within the specified region.

Stimulus: User adjusts sensitivity parameters for tree identification. Response: System fine-tunes its analysis based on user input.

Functional Requirements

REQ-1: The system shall provide an interface for users to upload and manage satellite images.

REQ-2: The system shall employ image processing algorithms to detect and enumerate trees within the defined forest area.

REQ-3: Users should be able to define and modify the geographical boundaries of the diversion area.

REQ-4: The system shall allow users to adjust sensitivity parameters to optimize tree identification accuracy.

REQ-5: In case of errors or inaccuracies, the system shall provide feedback to users and suggest corrective actions.

REQ-6: The system shall generate a comprehensive report containing the tree enumeration results.

REQ-7: Data confidentiality and security measures shall be implemented to protect sensitive information related to forest land.

3.4.2 System Feature 2

Feature Name: Machine Learning Model Training for Tree Species Classification

Description and Priority Description: This feature involves training a machine learning model to classify tree species based on image data. The system will utilize labeled image datasets to develop a classification algorithm capable of identifying trees present within forest areas.

Priority: Medium

Priority Components:

Benefit: 7

Penalty: 3

Cost: 6

Risk: 6

Stimulus/Response Sequences Stimulus: User provides labeled image datasets containing tree

species data. Response: System preprocesses and augments the data for training.

Stimulus: User selects the machine learning algorithm and parameters for model training.

Response: System trains the model using the provided datasets and specified parameters.

Stimulus: User evaluates the trained model's performance. Response: System provides metrics and visualizations to assess the model's accuracy and effectiveness.

Functional Requirements

REQ-1: The system shall support the ingestion and management of labeled image datasets for training.

REQ-2: Various machine learning algorithms shall be available for model training, including but not limited to convolutional neural networks (CNNs) and random forests.

REQ-3: Users should be able to specify hyperparameters and optimization techniques for model training.

REQ-4: The system shall provide tools for data preprocessing, including normalization, resizing, and augmentation.

REQ-5: Users should have the ability to fine-tune the trained model based on feedback and additional data.

REQ-6: Documentation and tutorials should be provided to assist users in understanding and utilizing the machine learning model training functionality

3.4.3 System Feature 3

Feature Name: Geospatial Mapping Integration for Land Cover Analysis

Description and Priority Description: This feature involves integrating geospatial mapping capabilities to analyze land cover within forest areas. The system will utilize satellite imagery and GIS techniques to classify land cover types, such as forested areas, agricultural land, water bodies, and settlements.

Priority: High

Priority Components:

Benefit: 9

Penalty: 2

Cost: 8

Risk: 4

Stimulus/Response Sequences Stimulus: User provides satellite imagery and defines the study area.

Response: System processes the imagery to generate a detailed land cover map.

Stimulus: User selects specific land cover classes for analysis. Response: System highlights and provides statistics on the distribution of selected land cover types within the study area.

Stimulus: User requests temporal analysis of land cover changes over time. Response: System generates time-series data and visualizations to illustrate land cover dynamics.

Functional Requirements

REQ-1: The system shall support the import and processing of satellite imagery in various formats, including raster datasets.

REQ-2: Geospatial analysis tools shall be available for land cover classification, including supervised and unsupervised classification methods.

REQ-3: Users should be able to define and customize land cover classification schemes based on specific project requirements.

REQ-4: The system shall provide tools for spatial analysis, including overlay operations, proximity analysis, and terrain modeling.

REQ-5: Users should have the ability to generate land cover maps with customizable legend and symbology options.

REQ-6: The system shall support multi-temporal analysis to track changes in land cover over different time periods.

REQ-7: Users should be able to export land cover analysis results in commonly used formats, such as GeoTIFF or shapefiles.

REQ-8: Documentation and training materials should be provided to assist users in utilizing geospatial mapping and land cover analysis functionality effectively.

3.5 Other Nonfunctional Requirements

3.5.1 Performance Requirements

1. Image Processing Speed:

Requirement: The system should process satellite images for tree enumeration within a maximum of 10 minutes per image. Rationale: Rapid processing is essential for timely decision-making regarding forest land diversion. Delays in image analysis could hinder the project's progress and effectiveness.

2. Machine Learning Model Inference Time:

Requirement: The trained machine learning model for tree species classification should provide inference results in less than 1 second per image. Rationale: Swift classification is necessary to

support real-time decision-making during land cover analysis. Longer inference times may lead to delays in obtaining actionable insights.

3. Geospatial Mapping Efficiency:

Requirement: Land cover mapping and analysis tasks should be completed within a maximum of 30 minutes for a study area of 100 square kilometers. Rationale: Efficiency in land cover analysis is crucial for promptly identifying areas of interest and assessing environmental changes. Delays in mapping could impede timely intervention and planning.

4. Scalability:

Requirement: The system should be able to handle datasets of up to 1 terabyte in size for image processing and analysis tasks. Rationale: As the project scales or when dealing with large geographic areas, the system must efficiently manage and process substantial volumes of data without performance degradation.

5. Accuracy and Reliability:

Requirement: The system's accuracy for tree enumeration, species classification, and land cover mapping should exceed 90%. Rationale: High accuracy is critical for reliable decision-making in forest management and land use planning. Inaccuracies could lead to misinterpretation of data and erroneous conclusions.

6. Resource Utilization:

Requirement: The system should utilize no more than 80% of available resources. Rationale: Efficient resource utilization ensures system stability and prevents performance bottlenecks that could degrade user experience and productivity.

7. Real-time Feedback:

Requirement: Users should receive real-time feedback on the progress of image processing, model training, and land cover analysis tasks. Rationale: Real-time feedback enhances user engagement and enables prompt intervention if issues arise during data processing or analysis.

3.5.2 Safety Requirements

1. Data Privacy and Confidentiality:

Requirement: The system must adhere to strict data privacy regulations, such as GDPR, HIPAA, or local data protection laws, ensuring the confidentiality of sensitive information related to forest land and landowners. Safeguard: Implement robust encryption mechanisms to protect data both at rest and in transit. Access controls should be enforced to restrict unauthorized access to sensitive data.

2. Safety of Personnel:

Requirement: Ensure that personnel involved in data collection or fieldwork activities are equipped with appropriate safety gear and trained in safety protocols to prevent accidents or injuries during field surveys. Safeguard: Conduct regular safety training sessions for personnel, emphasizing the importance of adhering to safety guidelines and procedures while working in forested areas.

Provide emergency communication devices for personnel to use in case of emergencies.

3. Environmental Impact:

Requirement: The system should minimize its environmental impact during data collection, processing, and analysis activities, taking into account factors such as energy consumption, waste generation, and ecological disturbance. Safeguard: Implement sustainable practices, such as using energy-efficient hardware, minimizing paper usage through digital documentation, and adhering to environmental regulations for data collection activities in sensitive ecosystems.

4. Compliance with Regulatory Standards:

Requirement: Ensure that the system complies with relevant regulatory standards and certifications related to environmental protection, forest management, and land use planning. Safeguard:

Regularly monitor updates to regulatory requirements and adapt the system's functionalities and workflows to remain compliant. Obtain necessary certifications or approvals from regulatory bodies to demonstrate adherence to safety and environmental standards.

3.5.3 Security Requirements

1. User Authentication and Access Control:

Requirement: The system must implement user authentication mechanisms to verify the identity of users accessing the system. Access control measures should be enforced to ensure that users only have access to data and functionalities appropriate for their roles and responsibilities.

Implementation: Utilize strong authentication methods, such as multi-factor authentication (MFA) or biometric authentication, to validate user identities. Implement role-based access control (RBAC) to manage user permissions and restrict access to sensitive data.

2. Data Encryption:

Requirement: All sensitive data, including user credentials, personally identifiable information (PII), and confidential forest land data, must be encrypted both at rest and in transit to prevent unauthorized access or interception. Implementation: Employ industry-standard encryption

algorithms, such as AES (Advanced Encryption Standard), to encrypt data stored in databases and data transmitted over networks. Use secure communication protocols, such as HTTPS, for data transmission to ensure encryption during transit.

3. Secure Data Storage and Handling:

Requirement: The system should adhere to secure data storage practices to protect against data

breaches or unauthorized access. Measures should be in place to prevent data loss, corruption, or tampering. Implementation: Store sensitive data in secure, encrypted databases with access controls and audit trails to monitor data access and modifications. Implement regular data backups and disaster recovery procedures to ensure data resilience in case of unexpected events.

4. Vulnerability Management:

Requirement: The system must have processes in place to identify, assess, and remediate security vulnerabilities in software components, libraries, and dependencies. Implementation: Establish a vulnerability management program to regularly scan the system for vulnerabilities using automated tools and manual assessments. Implement patch management procedures to apply security patches and updates promptly.

5. Logging and Auditing:

Requirement: The system should maintain detailed logs of user activities, system events, and access attempts to facilitate security monitoring, incident response, and compliance with regulatory requirements. Implementation: Enable comprehensive logging of system activities, including login attempts, data access, and configuration changes. Implement log management solutions to centralize and analyze log data for detecting security incidents and investigating suspicious activities.

6. Regulatory Compliance:

Requirement: Ensure that the system complies with relevant security and privacy regulations, such as GDPR, CCPA, or industry-specific standards like ISO 27001. Implementation: Conduct regular audits and assessments to verify compliance with regulatory requirements. Obtain necessary certifications, such as SOC 2 or ISO 27001, to demonstrate adherence to security and privacy standards.

3.5.4 Software Quality Attributes

1. Reliability:

Requirement: The system should maintain a system uptime of at least 99.9%. Verifiable: Monitor system uptime and downtime using automated monitoring tools, with periodic reports on availability metrics.

2. Maintainability:

Requirement: The system codebase should adhere to coding standards and best practices, with clear documentation and modular design to facilitate ease of maintenance and future enhancements.

Verifiable: Conduct code reviews and static code analysis to ensure adherence to coding standards,

and track maintenance efforts over time to assess maintainability improvements.

3. Usability:

Requirement: The user interface (UI) should be intuitive and user-friendly, with clear navigation and well-defined workflows to minimize user errors and facilitate task completion. Verifiable:

Conduct usability testing with representative users to gather feedback on the UI design and user experience, and measure task completion rates and user satisfaction scores to assess usability.

4. Performance Efficiency:

Requirement: The system should be capable of handling concurrent user requests and processing large datasets efficiently, with response times averaging less than 1 second for critical operations.

Verifiable: Conduct performance testing under various load conditions to measure response times and system throughput, and optimize system resources to achieve desired performance levels.

5. Security:

Requirement: Implement robust security measures, including encryption, access control, and vulnerability management, to protect against unauthorized access, data breaches, and cyber threats.

Verifiable: Conduct regular security audits and penetration testing to identify and remediate security vulnerabilities, and track compliance with security standards and regulations through audit logs and security assessments.

3.5.5 Business Rules

1. Administrative Access Control:

Only users with administrative privileges, such as system administrators or designated managers, are allowed to access and modify system configuration settings, user roles, and permissions.

2. Data Ownership and Access Rights:

Users are only permitted to access and manipulate data within their designated scope of authority. Access to sensitive or restricted data, such as land ownership records or environmental impact assessments, is limited to authorized personnel.

3. Approval Workflow for Land Diversion Requests:

Before approving any requests for forest land diversion, designated authorities, such as government agencies or environmental review boards, must review and evaluate the associated environmental impact assessments, land use plans, and compliance with regulatory requirements.

4. Compliance with Regulatory Standards:

All activities conducted within the system must adhere to applicable laws, regulations, and industry standards governing forest management, environmental protection, and land use planning.

Noncompliance may result in penalties, fines, or legal repercussions.

5. Confidentiality of Landowner Information:

Personal information and sensitive data belonging to landowners, such as contact details, property boundaries, and land use agreements, must be treated with strict confidentiality and only shared with authorized parties on a need-to-know basis.

6. Transparency and Accountability:

All decisions and actions taken within the system, including land diversion approvals, data modifications, and audit trail entries, must be documented and traceable to ensure transparency and accountability in decision-making processes.

7. Environmental Conservation Prioritization:

In cases where forest land diversion is considered, priority should be given to preserving ecologically sensitive areas, biodiversity hotspots, and habitats of endangered species, in accordance with conservation priorities and ecological principles.

8. Stakeholder Consultation and Engagement:

Before initiating any major land use changes or development projects, stakeholders, including local communities, indigenous groups, and environmental organizations, should be consulted and provided with opportunities for meaningful engagement and participation in decision-making processes.

3.6 Other Requirements

1. Database Requirements:

The system shall utilize a relational database management system (RDBMS) to store and manage structured data, such as user profiles, image metadata, and land cover classifications. Database backups shall be performed regularly to ensure data integrity and facilitate disaster recovery.

2. Internationalization Requirements:

The user interface (UI) shall support multiple languages to accommodate users from diverse linguistic backgrounds. Date and time formats, currency symbols, and measurement units shall be configurable based on user preferences or regional standards.

3. Legal Requirements:

The system shall comply with all relevant laws, regulations, and legal frameworks governing forest management, environmental conservation, and land use planning in the project's geographical jurisdiction. Legal disclaimers and terms of service shall be provided to users, outlining their rights and responsibilities when using the system.

4. Reuse Objectives:

Where applicable, the system shall incorporate reusable components, libraries, or frameworks to streamline development efforts, promote code maintainability, and facilitate future enhancements. Code documentation and modular design principles shall be followed to encourage code reuse and extensibility.

Chapter 4

System Design

4.1 Introduction to System Design

The System Design for Image Analytics for Tree Enumeration for Diversion of Forest Land employs advanced image processing, machine learning, and spatial analysis techniques to analyze high-resolution satellite or aerial images. It systematically processes data through stages including preprocessing, segmentation, feature extraction, object detection, enumeration, integration, analysis, validation, and reporting. This design emphasizes scalability, flexibility, and ethical considerations, ensuring the system can handle large datasets, adapt to technological advancements, and adhere to ethical standards. By providing accurate tree enumeration data and actionable insights, the system supports informed decision-making for sustainable land management and conservation in forest land diversion projects.

4.2 System Architecture

- The System Architecture for Image Analytics for Tree Enumeration for Diversion of Forest Land encompasses data acquisition, preprocessing, image processing, and analysis layers. It integrates advanced algorithms for segmentation, feature extraction, and object detection to accurately enumerate trees within forested areas. The architecture also includes data integration, validation, and reporting components for comprehensive analysis and decision support. Scalability and flexibility are key design considerations, ensuring the system can handle large datasets and adapt to evolving technologies. Ethical considerations such as data privacy and responsible use of technology are incorporated to maintain integrity in environmental monitoring and land management practices.

Appendix A: Glossary

RDBMS: Relational Database Management System, a software system used to manage and

organize relational databases.

UI: User Interface, the visual and interactive components of a software application that users interact with.

Appendix B: Analysis Models

Activity Diagram:

Activity diagrams are a type of behavior diagram in Unified Modeling Language (UML) that depict the flow of control or the sequence of activities within a system. They provide a visual representation of the dynamic behavior of a system, focusing on the workflow and the interactions between various components or entities. Activity diagrams are widely used in software engineering and business process modeling to describe complex processes, workflows, use cases, and system behaviors in a clear and intuitive manner.

In an activity diagram, activities are represented as rounded rectangles, while transitions between activities are depicted as arrows. Decisions and branching points are shown using diamond-shaped decision nodes, allowing for the representation of conditional flows within the process. Fork and join nodes are used to indicate parallel or concurrent activities, enabling the modeling of concurrent behavior within the system.

Activity diagrams can be used at different levels of abstraction, from high-level overviews of system processes to detailed specifications of individual use cases or scenarios. They help stakeholders and developers to understand the sequence of actions, decision points, and control flows within the system, facilitating communication and collaboration among project teams.

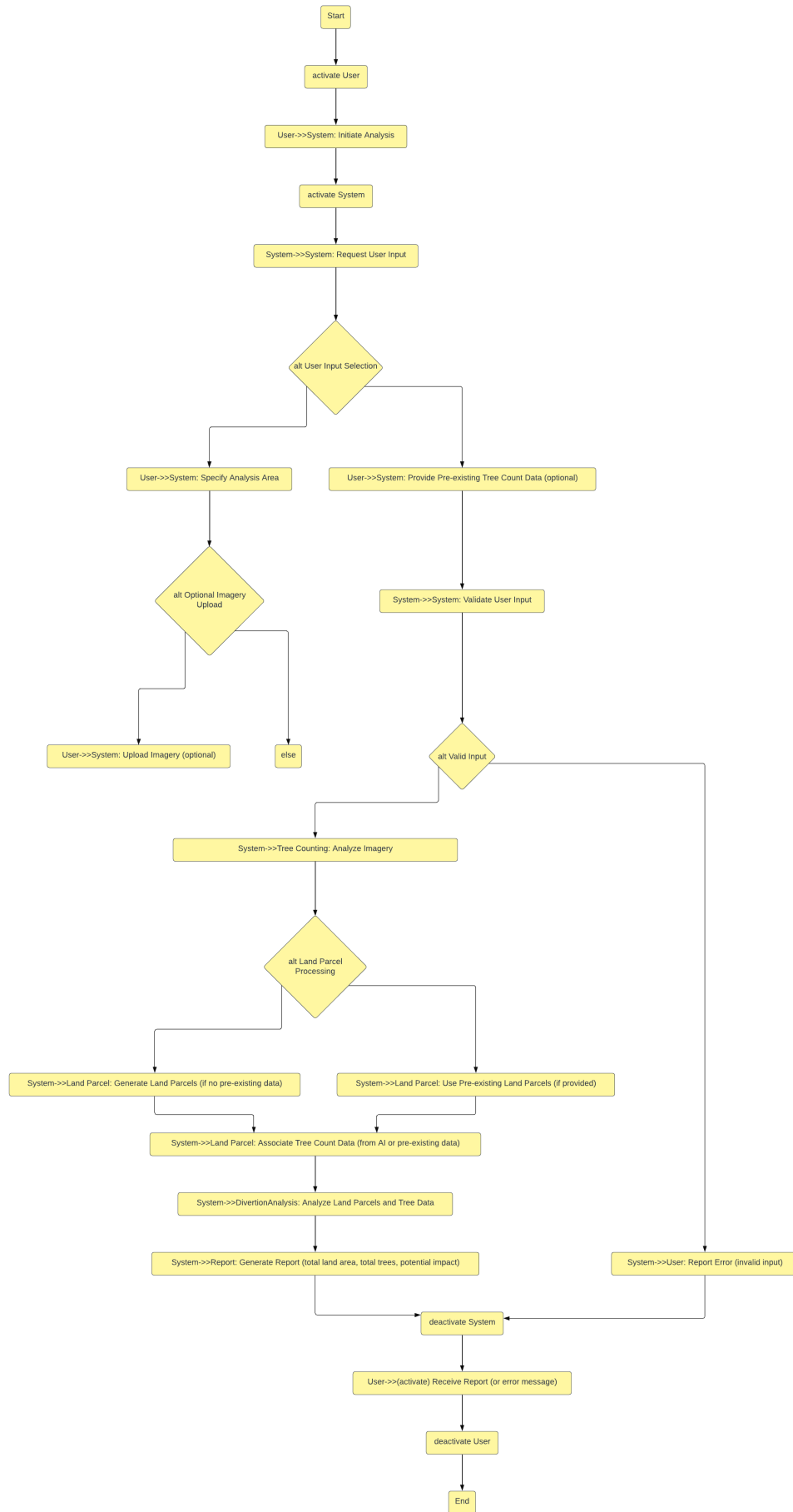


Figure 4.1: Activity Diagram

Use Case Diagram

Use case diagrams are a type of behavioral diagram in the Unified Modeling Language (UML) used to visually represent the interactions between actors (users or external systems) and a system under consideration. They help in understanding the functionality of a system from a user's perspective.

Here are some key concepts and components of a use case diagram:

Actors:

Actors represent the roles that interact with the system. They could be users, other systems, or any external entities. Actors are depicted as stick figures outside the system boundary.

Use Cases:

Use cases represent the specific functionalities or tasks that the system provides to its actors.

They describe the behavior of the system from the perspective of its users.

Use cases are depicted as ovals inside the system boundary. Relationships:

Associations: Represent relationships between actors and use cases, indicating that an actor is involved in a particular use case.

Include: Indicates that one use case includes the functionality of another. It's used when one use case (the base) includes the behavior of another use case (the included).

Extend: Indicates that one use case extends another. It's used when a use case (the extension) adds new behavior to another use case (the extended) under certain conditions.

Generalization: Indicates an inheritance relationship between use cases. It's used when one use case is a specialized version of another.

System Boundary:

Represents the scope or boundary of the system under consideration.

It encloses all the use cases of the system. Associations:

Associations connect actors with use cases, representing the interactions between them.

Associations can have names and multiplicities to indicate how many instances of each actor participate in a use case.

When creating a use case diagram, it's essential to focus on capturing the primary functionalities of the system and the actors that interact with it. It should provide a high-level overview of the system's behavior without delving into implementation details.

Sequence Diagram:

Sequence diagrams are a type of UML (Unified Modeling Language) diagram used to illustrate the

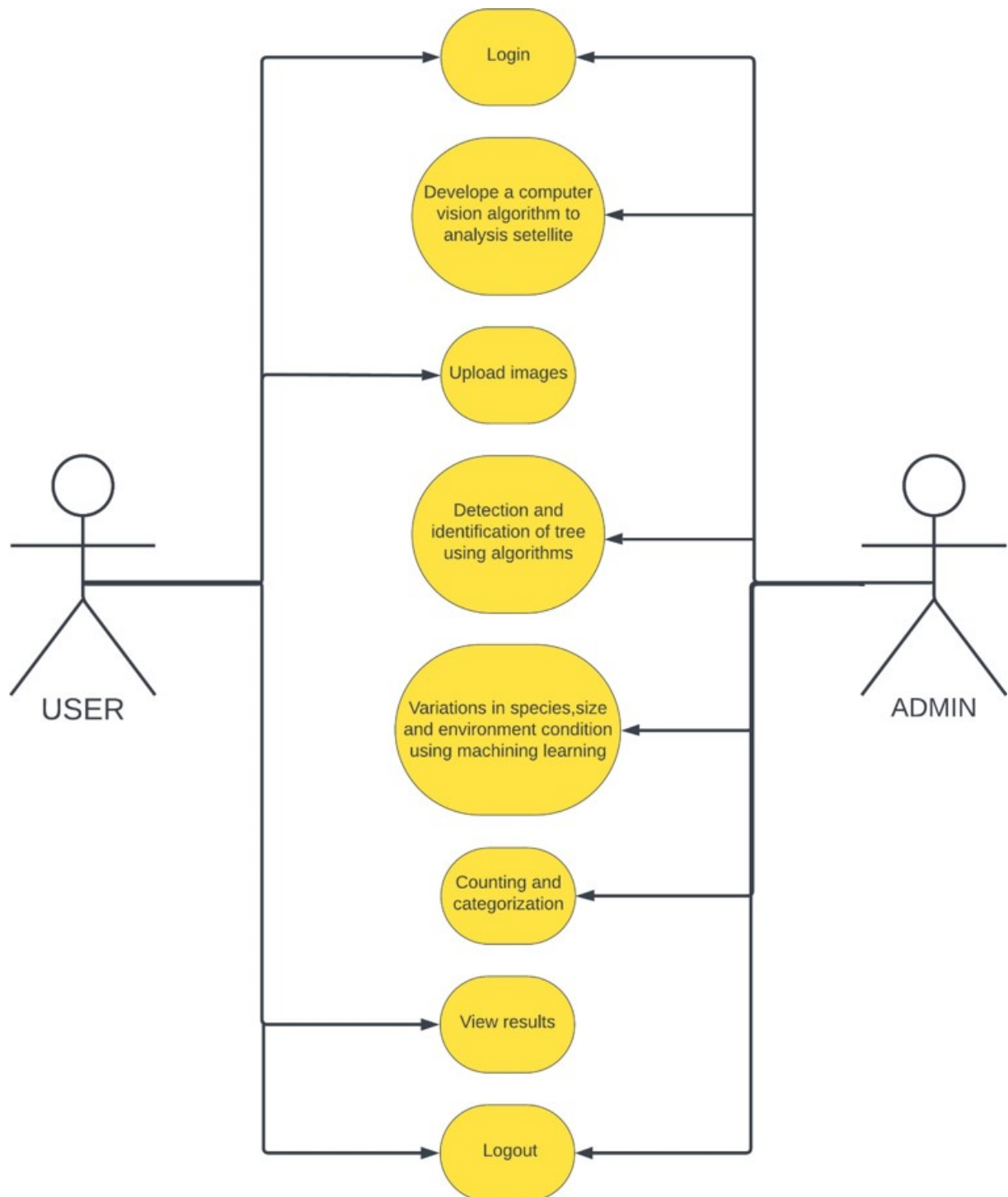


Figure 4.2: Use Case Diagram

interactions between objects within a system over a specific period, typically representing a particular scenario or use case. At their core, sequence diagrams visualize the sequence of messages exchanged between objects as they collaborate to accomplish a task or achieve a particular behavior. Objects, which represent instances of classes or components in the system, are depicted as rectangular boxes with their names. These objects interact by sending messages to each other, with messages displayed as arrows between lifelines, which represent the existence of an object over time. Lifelines are vertical dashed lines extending from the top of an object box downward, showing the timeline of an object's interactions.

Messages can be synchronous, indicating that the sender waits for a response from the receiver, or asynchronous, where no immediate response is expected. Return messages, denoted by dashed lines with an arrowhead pointing back to the sender, represent responses or return values sent by the receiver object. Activation boxes, depicted as rectangles on lifelines, show the period during which an object is performing an operation, illustrating the duration of its activity or processing time within the sequence.

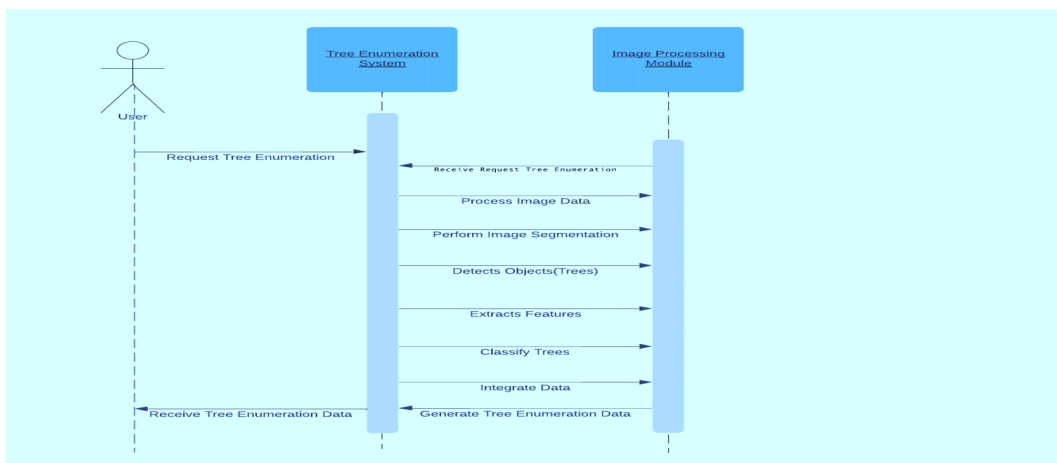


Figure 4.3: Sequence Diagram

Collaboration Diagram:

A collaboration diagram, also known as a communication diagram, is a type of diagram used in Unified Modeling Language (UML) to visualize the interactions and relationships between objects or components within a system. It emphasizes the structural organization of the system and the messages exchanged between the objects during their collaboration.

Collaboration diagrams are particularly useful for visualizing the runtime behavior of a system, illustrating how objects interact with each other to accomplish a particular task or scenario. They help in understanding the flow of control and information within the system and identifying potential communication issues or bottlenecks.

When creating collaboration diagrams, it's essential to focus on clarity and simplicity to effectively convey the intended interactions and relationships between objects. Annotations and notes can be used to provide additional context or explanations where necessary.

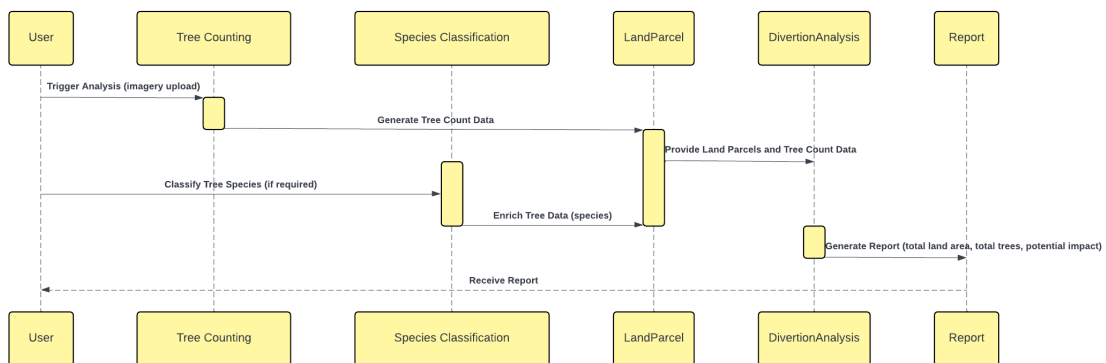


Figure 4.4: Collaboration Diagram

Class Diagrams:

Class diagrams depict the structure of classes and their relationships in object-oriented programming (OOP), illustrating the attributes, methods, and associations of each class. They provide a blueprint for the system's object-oriented design, including inheritance, encapsulation, and polymorphism.

Class diagrams help in designing and modeling the system's architecture, identifying key classes, their properties, and interactions. They facilitate communication between developers and stakeholders, ensuring a common understanding of the system's design principles and components.

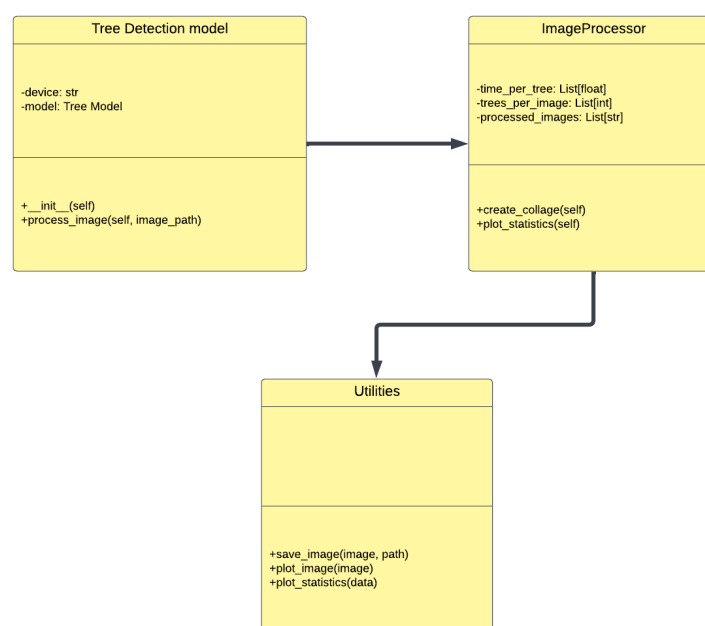


Figure 4.5: Class Diagram

Object Diagram:

An object diagram is a type of structural diagram in Unified Modeling Language (UML) that provides a snapshot of the objects and their relationships within a system at a specific point in time. It emphasizes the static structure of the system by depicting the instances of classes or objects and the links between them. Object diagrams are particularly useful for visualizing concrete instances of classes and their associations, aggregations, or compositions.

In an object diagram, objects are represented as rectangles with the object's name and attributes displayed inside. Links between objects, which represent relationships or associations, are depicted using lines connecting the objects. Unlike class diagrams, which represent the structure of classes and their relationships in a more abstract and generalized manner, object diagrams focus on the specific instances of these classes and their current relationships.

Object diagrams are invaluable during the design and development phases of a system as they provide a tangible representation of the system's structure, aiding in the identification of object instances, their attributes, and their relationships. They can also be used to validate the correctness of class designs and relationships by visually inspecting concrete examples of objects and their interactions.

Furthermore, object diagrams can complement other UML diagrams such as class diagrams, sequence diagrams, and use case diagrams, providing different perspectives on the system's structure and behavior. They are particularly useful for communication between stakeholders, designers, and developers, as they offer a concrete visualization of the system's internal structure and relationships.

Appendix C: To Be Determined List

TBD-1: Define specific database schema and relationships, including tables, fields, and their associations.

TBD-2: Identify language options and implement internationalization features, such as language selection and localization of UI elements.

TBD-3: Determine legal requirements and implications for system design and functionality, including compliance with data protection regulations and intellectual property laws.

TBD-4: Specify backup and disaster recovery procedures for the database, including backup frequency, storage locations, and restoration processes.

TBD-5: Define the scope and format of legal disclaimers, terms of service, and privacy policies to be included in the system documentation and user interfaces.

TBD-6: Determine the extent of code reuse and identify reusable components, libraries, or frameworks to be integrated into the system architecture.

TBD-7: Clarify user roles and permissions, including administrative privileges and access levels for different system functionalities.

TBD-8: Specify performance benchmarks and thresholds for key system operations, such as image processing speed and model inference times.

TBD-9: Identify external dependencies and integration points with third-party services or APIs, including mapping services and satellite imagery providers.

TBD-10: Define testing strategies and methodologies, including unit testing, integration testing, and user acceptance testing, to ensure the reliability and quality of the system.

4.3 Functional Components

- Functional Components for Image Analytics for Tree Enumeration for Diversion of Forest Land include data acquisition for images and environmental data, preprocessing for format standardization and quality enhancement, segmentation for tree identification, feature extraction for canopy size and color analysis, object detection using algorithms like Faster R-CNN, tree enumeration for accurate counts, data integration for analysis, visualization for insights, validation processes for result accuracy, and reporting for decision support. These components collectively enable systematic processing, analysis, and reporting of image data to support informed decision-making in forest land diversion projects.

4.4 Technologies and Frameworks

- For Image Analytics for Tree Enumeration for Diversion of Forest Land, the technology stack includes OpenCV and PIL for image processing, TensorFlow or PyTorch for deep learning-based object detection like Faster R-CNN, GDAL and Fiona for geospatial data handling, NumPy/pandas and Matplotlib/Seaborn for analysis and visualization, QGIS or ArcGIS for GIS tasks, cloud platforms like AWS or Google Cloud for scalability, Git/GitHub for version control, Python/R for programming, and ethical tools like Fairlearn and Responsible AI toolkits for bias mitigation. These technologies collectively form a powerful framework for accurate tree enumeration and informed decision-making in forest land diversion projects.

4.5 Dataset

Raw data is a good place to start, but you obviously cannot just shove it into a machine learning algorithm and hope it offers you valuable insights into your customers' behaviors. There are quite a few steps you need to take before your dataset becomes usable.

4.5.1 Annotation

Annotation is likely the most important part of machine learning projects. Fancy models are nice, but data is always paramount. If you aren't happy with model performance, annotating new samples is always the best first idea.



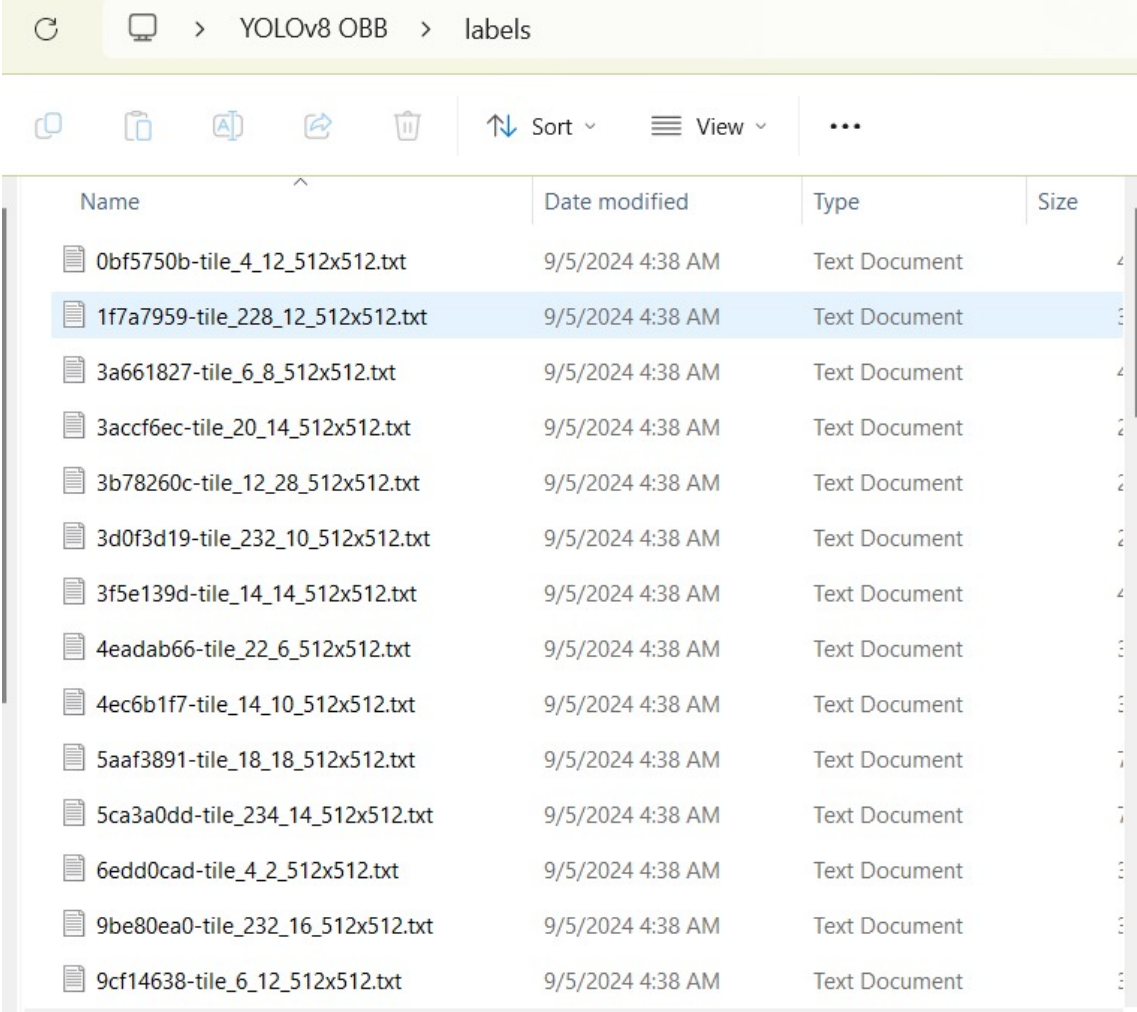
Figure 4.6: Satellite Image



Figure 4.7: Annotated Image

object detection models use the non-annotated areas of an image as negative data. We know that it can be exceptionally hard to annotate all trees in an image, or determine the classes of all birds in an image. However, if you have objects in the image that are not annotated, the model is learning to ignore those portion of the image. This can severely affect model performance.

Our Model is train on over 1500 images of Indian satellite map.



The screenshot shows a file explorer window with the address bar indicating the path: > YOLOv8 OBB > labels. The toolbar includes icons for refresh, back, forward, copy, paste, delete, sort, view, and a menu. The file list is as follows:

Name	Date modified	Type	Size
0bf5750b-tile_4_12_512x512.txt	9/5/2024 4:38 AM	Text Document	4
1f7a7959-tile_228_12_512x512.txt	9/5/2024 4:38 AM	Text Document	3
3a661827-tile_6_8_512x512.txt	9/5/2024 4:38 AM	Text Document	4
3accf6ec-tile_20_14_512x512.txt	9/5/2024 4:38 AM	Text Document	2
3b78260c-tile_12_28_512x512.txt	9/5/2024 4:38 AM	Text Document	2
3d0f3d19-tile_232_10_512x512.txt	9/5/2024 4:38 AM	Text Document	2
3f5e139d-tile_14_14_512x512.txt	9/5/2024 4:38 AM	Text Document	4
4eadab66-tile_22_6_512x512.txt	9/5/2024 4:38 AM	Text Document	3
4ec6b1f7-tile_14_10_512x512.txt	9/5/2024 4:38 AM	Text Document	3
5aaf3891-tile_18_18_512x512.txt	9/5/2024 4:38 AM	Text Document	7
5ca3a0dd-tile_234_14_512x512.txt	9/5/2024 4:38 AM	Text Document	7
6edd0cad-tile_4_2_512x512.txt	9/5/2024 4:38 AM	Text Document	3
9be80ea0-tile_232_16_512x512.txt	9/5/2024 4:38 AM	Text Document	3
9cf14638-tile_6_12_512x512.txt	9/5/2024 4:38 AM	Text Document	3

Figure 4.10: labels

Training Parameters

1. Dataset

Description of the tree detection dataset

data : tree_detection.yaml

2. Model Architecture

YOLOv8n-OBB architecture

Pretrained model utilization

model: yolov8n-obb.pt

3. Training Configuration

•Hyperparameters

epochs: 150

time: null

patience: 50

batch: 50

imgsz: 512

•Optimization strategy

optimizer: auto

•Data augmentation techniques

$hsv_h : 0.015$

$hsv_s : 0.7$

$hsv_v : 0.4$

degrees: 0.0

translate: 0.1

scale: 0.5

shear: 0.0

perspective: 0.0

flipud: 0.0

fliplr: 0.5

bgr: 0.0

mosaic: 1.0

mixup: 0.0

$copy_{paste} : 0.0$

autoaugment : *randaugment*

erasing: 0.4

cropfraction : 1.0

4. Hardware and Performance Optimization

- CUDA utilization

- Automatic Mixed Precision (AMP)

device: cuda

workers: 8

Chapter 5

Methodology

5.1 Image Acquisition

Acquire high-resolution satellite or aerial images covering the entire forest land area of interest. Ensure the images are captured with sufficient detail and clarity to facilitate accurate tree identification and analysis.

5.2 Pre-processing

Prior to analysis, preprocess the acquired images by converting them to a standardized format such as JPEG or PNG. Apply image enhancement techniques such as contrast adjustment, histogram equalization, and noise reduction to improve image quality. Additionally, perform geometric correction and orthorectification to rectify any distortions caused by terrain or camera angles, ensuring the images are suitable for further analysis.

5.3 Image Segmentation

Employ advanced image segmentation algorithms to partition the preprocessed images into distinct regions corresponding to trees and non-tree areas. Utilize segmentation criteria such as color, texture, shape, and spatial information to accurately delineate tree boundaries within the images.

5.4 Feature Extraction

Extract relevant features from the segmented tree regions to characterize individual trees. Features may include tree canopy size, shape descriptors, color attributes, texture patterns, and spatial distribution metrics. Apply feature extraction techniques such as morphological operations, edge detection, and texture analysis to capture key tree attributes for subsequent analysis.

5.5 Object Detection

Utilize state-of-the-art object detection algorithms, such as Faster R-CNN (Region-based Convolutional Neural Network) or YOLO (You Only Look Once), to detect and localize individual trees within the segmented regions. Train the object detection models using annotated data to accurately identify tree instances and differentiate them from other objects or background elements in the images.

5.6 Tree Enumeration

Implement tree enumeration algorithms to count the detected tree instances within the forest land area. Refine the enumeration results using post-processing techniques such as non-maximum suppression to eliminate duplicate detections and thresholding to filter out false positives, ensuring an accurate count of the actual trees present in the images.

5.7 Data Analysis

Conduct comprehensive data analysis on the enumerated trees to derive meaningful insights into tree density, spatial distribution patterns, species composition, and overall forest health. Utilize statistical analysis, data visualization techniques, and GIS (Geographic Information System) tools to analyze and interpret the tree enumeration data effectively.

5.8 Validation and Accuracy

Validate the accuracy and reliability of the tree enumeration results through rigorous validation processes, including ground truthing and comparison with known reference data. Calculate accuracy metrics such as precision, recall, and F1-score to assess the performance of the image analytics methodology and ensure the validity of the obtained results.

5.9 Decision Support

Compile the analyzed data, including tree counts, species information, accuracy assessments, and analytical insights, into comprehensive reports and decision support systems. Provide decision-makers, environmental agencies, and land managers with actionable information and recommendations to support informed decision-making regarding forest land diversion projects, conservation efforts, and sustainable land management practices.

Chapter 6

Implementation

Objective

To download satellite images of a specific area defined by latitude and longitude coordinates and analyze the images to detect and count trees using a machine learning model.

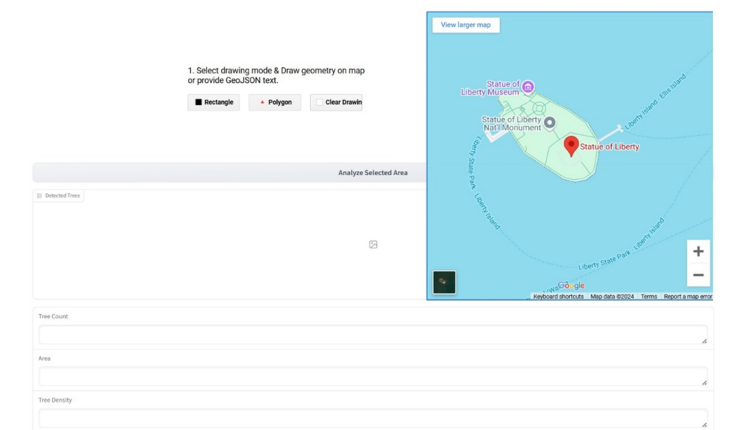


Figure 6.1: Objective

Code Structure

1. Image Downloading

```
def download_satellite_image_area(lat1, lon1, lat2, lon2, zoom):
    x1 = math.floor((lon1 + 180) / 360 * (1 << zoom))
    y1 = math.floor((1 - math.log(math.tan(math.radians(lat1)) + 1 / math.cos(math.radians(lat1))) / math.pi) / 2 * (1 << zoom))
    x2 = math.floor((lon2 + 180) / 360 * (1 << zoom))
    y2 = math.floor((1 - math.log(math.tan(math.radians(lat2)) + 1 / math.cos(math.radians(lat2))) / math.pi) / 2 * (1 << zoom))

    num_tiles_x = x2 - x1 + 1
    num_tiles_y = y2 - y1 + 1
```

Figure 6.2: Image Downloading

Calculate image tile indices and download each tile

Stitch tiles together into a single image

2. Image Processing

Load image and apply Deep Forest model for tree detection

```
def process_image(image_path):
    img_array = model.predict_image(path=image_path, return_plot=True)

    img = Image.fromarray(img_array.astype('uint8'), 'RGB')

    output_path = os.path.join(output_dir, f"detected_{os.path.basename(image_path)}")
    img.save(output_path)

    return output_path

def process_satellite_image_area(lat1, lon1, lat2, lon2, zoom):
    image_path = download_satellite_image_area(lat1, lon1, lat2, lon2, zoom)
    if image_path:
        return process_image(image_path)
    return None
```

Figure 6.3: Image Processing

Save and return the processed image

3. Gradio Interface

```
iface = gr.Interface(
    fn=process_satellite_image_area,
    inputs=[
        gr.Number(label="Start Latitude", value=40.7128),
        gr.Number(label="Start Longitude", value=-74.0060),
        gr.Number(label="End Latitude", value=40.7138),
        gr.Number(label="End Longitude", value=-74.0050),
        gr.Slider(minimum=1, maximum=20, step=1, label="Zoom Level", value=18)
    ],
    outputs=gr.Image(label="Detected Trees",
        title="Tree Detection using DeepForest",
        description="Enter start and end coordinates to define an area, and zoom level to download and process a satellite image fo
    )
```

Figure 6.4: Gradio Interface

When this code is run, it creates a web interface where users can:

Enter the starting and ending coordinates of a rectangular area on the Earth's surface.

Select the desired zoom level for the satellite image.

Accuracy Tests:

1 Mean Avg Precision

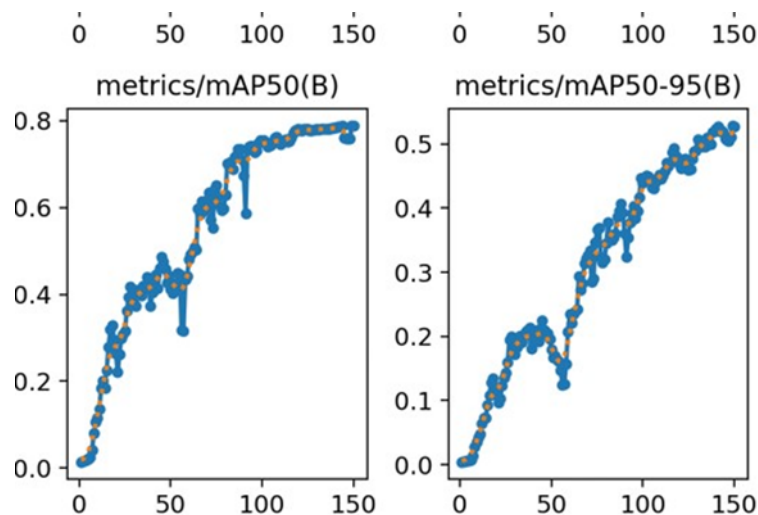


Figure 6.5: Mean Avg Precision

The image shows two line graphs representing the performance of a model on two different metrics: mAP50(B) and mAP50-95(B). The x-axis of both graphs represents the number of training epochs, while the y-axis represents the metric value. The graphs show that the model's performance on both metrics improves as the number of training epochs increases, with mAP50(B) reaching a plateau at

around 100 epochs and mAP50-95(B) continuing to improve slightly beyond 100 epochs.

- Evaluate mAP (mean Average Precision) at IoU threshold of (0.7). iou: 0.7
- Test model performance with different confidence thresholds
- Assess performance on various tree sizes and densities

2 Precision-Recall curve

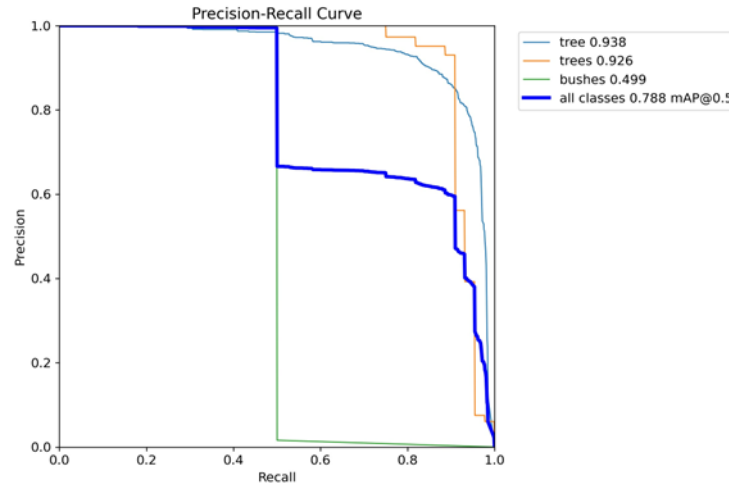


Figure 6.6: Precision-Recall curve

The image shows a precision-recall curve for a multi-class classification model. The x-axis represents recall, which measures the proportion of true positive instances that were correctly identified, while the y-axis represents precision, which measures the proportion of instances predicted as positive that were actually positive. The curve shows the trade-off between precision and recall for different classification thresholds. The blue line represents the overall performance of the model for all classes, while the other lines represent the performance for individual classes (tree, trees, bushes). The mAP@0.5 value of 0.788 indicates the average precision for all classes at a recall threshold of 0.5.

3 F1-Confidence curve

The image shows an F1-confidence curve for a multi-class classification model. The x-axis represents confidence, which is a measure of the model's certainty in its prediction, while the y-axis represents F1-score, which is a harmonic mean of precision and recall. The curve shows the trade-off between F1-score and confidence for different classification thresholds. The blue line represents the overall performance of the model for all classes, while the other lines represent the performance for individual classes (tree, trees, bushes). The "all classes 0.78 at 0.289" label indicates that the model achieves an F1-score of 0.78 when using a confidence threshold of 0.289.

Oriented Bounding Box(OBB) Specific Test:

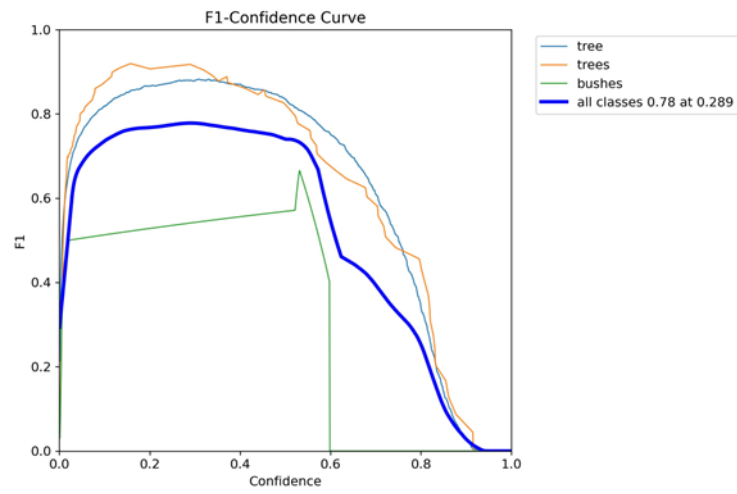


Figure 6.7: F1-Confidence curve

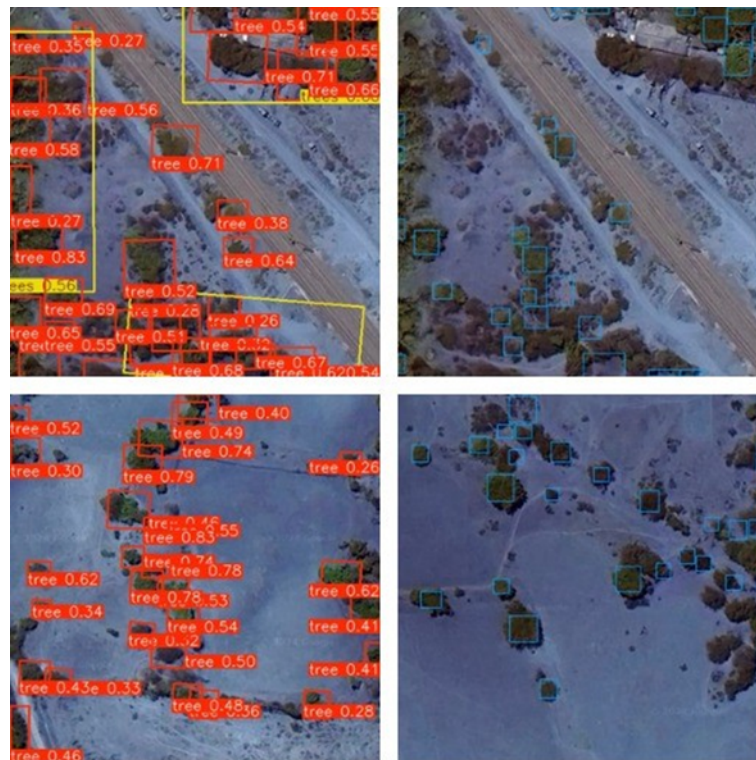


Figure 6.8: Oriented Bounding Box(OBB) Specific Test

- Evaluate accuracy of orientation prediction for trees
- Test performance on trees with different aspect ratios

1. Data Augmentation Effectiveness:

- Compare model performance with and without the following augmentations: hsv h: 0.015 hsv s: 0.7 hsv v: 0.4 degrees: 0.0 translate: 0.1 scale: 0.5 shear: 0.0 perspective: 0.0 flipud: 0.0 fliplr: 0.5 bgr: 0.0 mosaic: 1.0 mixup: 0.0 copy paste: 0.0 auto augment: randaugment erasing: 0.4 crop fraction: 1.0

2. Hardware Utilization:

- Measure GPU memory usage during inference
- Evaluate inference speed (FPS) on CUDA device device: cuda

3. Model Robustness:

- Test on images with varying lighting conditions (using HSV augmentation parameters)
- Evaluate performance on partially occluded trees
- Assess accuracy on images with different resolutions (scaled up/down from 512x512)imgsz: 512

4. Detection Limits:

- Evaluate the model's performance with the maximum detection limit of 300 objects per image max det: 300

5. False Positive/Negative Analysis:

- Test on images with no trees to check for false positives
- Evaluate on densely forested areas to check for false negatives

6. Model Generalization:

- Test on different types of aerial imagery (if available) to assess generalization
- Evaluate performance on images taken at different times of year (seasonal variations)

7. Edge Case Handling:

- Test on images with trees at image boundaries
- Evaluate performance on images with extreme tree densities (very sparse and very dense)These test cases cover a wide range of scenarios and will help in thoroughly evaluating the model's performance for the tree detection task using oriented bounding boxes.

8. Comparison Tests:

- Compare performance with other OBB models or previous versions of YOLOv8-OBB
- Benchmark against traditional (non-OBB) object detection models for tree detection

6.0.1 Testing

Test Case 1: Model Output for Single Image

Objective: Verify that the model generates bounding boxes for a single input image.

Steps:

Load a sample image from the validation set. Run the image through the trained model. Inspect the output for bounding boxes and associated labels.

Expected Result:

The model should output bounding boxes and labels for objects detected in the image. The bounding boxes should be in the format specified by the OBB (Object Bounding Box) detection.

Test Case 2: Model Performance Metrics

Objective: Ensure the model meets the performance metrics defined during training (e.g., accuracy, precision, recall).

Steps:

Evaluate the model on the validation dataset.

Calculate performance metrics such as precision, recall, and F1 score.

Compare the metrics against the expected benchmarks or thresholds.

Expected Result:

Performance metrics should meet or exceed the predefined benchmarks or thresholds.

Test Case 3: Handling Overlapping Objects

Objective: Test the model's ability to correctly detect and handle overlapping objects.

Steps:

Load images from the validation set that contain overlapping objects.

Run these images through the model.

Check if the model correctly detects and distinguishes between overlapping objects.

Expected Result:

The model should successfully identify and differentiate overlapping objects.

Test Case 4: Label Accuracy

Objective: Validate that the model's output labels are accurate.

Steps:

Run a set of validation images through the model.

Compare the predicted labels with the ground truth labels.

Calculate the label accuracy.

Expected Result:

The predicted labels should closely match the ground truth labels, indicating accurate label detection.

Test Case 5: Bounding Box Precision

Objective: Ensure the precision of the bounding boxes predicted by the model.

Steps:

Run validation images through the model.

Compare the predicted bounding boxes to the ground truth bounding boxes.

Calculate the Intersection over Union (IoU) for each bounding box.

Expected Result:

The IoU should be above a certain threshold (e.g., 0.5) for the bounding boxes to be considered precise.

Test Case 6: Model Inference Time

Objective: Verify that the model's inference time is within acceptable limits.

Steps:

Measure the time taken to run a batch of images through the model.

Compare the inference time against acceptable performance benchmarks.

Expected Result:

The inference time should be within the acceptable limits, ensuring efficient model performance.

Test Case 7: Robustness to Different Lighting Conditions

Objective: Test the model's robustness to varying lighting conditions.

Steps:

Prepare a set of validation images with different lighting conditions.

Run these images through the model.

Evaluate the model's performance and detection accuracy under these varied conditions.

Expected Result:

The model should maintain good detection performance across different lighting conditions.

Test Case 8: Model Stability Across Epochs

Objective: Ensure that the model's performance is stable across different epochs.

Steps:

Evaluate the model's performance at various checkpoints (e.g., every 10 epochs). Compare the performance metrics over time.

Expected Result:

The model should show consistent or improving performance metrics across epochs.

Test Case 9: Handling Edge Cases

Objective: Evaluate how the model handles edge cases such as very small or very large objects.

Steps:

Test the model with images containing very small or very large objects.

Analyze the model's ability to detect and accurately bound these edge cases.

Expected Result:

The model should be able to detect and correctly bound very small or very large objects.

Test Case 10: Model Compatibility

Objective: Confirm that the trained model can be loaded and used in different environments.

Steps:

Save the trained model.

Load the model in a different environment or system.

Perform inference on a sample image.

Expected Result:

The model should load successfully and perform inference as expected in the new environment.

Chapter 7

Conclusion

7.1 Project Conclusion

Tree Detection Automation Leveraging Satellite Imagery project effectively counted trees, analyzed species, and mapped spatial distributions, aiding sustainable land use decisions. Validations ensured result accuracy, contributing to environmental impact assessments and informed development practices. Future enhancements might involve real-time monitoring and automated reporting for improved efficiency. The project's comprehensive reports and data-driven insights facilitate ecological preservation while supporting responsible land management. Overall, this project showcases the potential of image analytics in environmental monitoring, emphasizing the importance of technology in balancing development with conservation efforts.

Chapter 8

Future Work

1. Improving Efficiency:

Optimization of Model Inference: Exploring more efficient algorithms for image processing and model inference to reduce the time required for tree detection. Techniques such as model quantization, pruning, and exploring more efficient model architectures could be beneficial.

Parallel Processing: Leveraging parallel processing techniques and hardware acceleration (e.g., GPUs, TPUs) to process multiple images simultaneously, thereby speeding up the analysis.

Caching Mechanisms: Implementing caching for model predictions, particularly for overlapping areas in large datasets, to avoid redundant computations and improve overall efficiency.

2. Enhancing User Functionality:

Interactive Visualization Tools: Developing web-based interfaces or interactive visualization tools that enable users to explore detected tree crowns, edit annotations, and review model predictions directly.

Integration with GIS Software: Creating plugins or extensions for popular Geographic Information System (GIS) software to allow direct import and analysis of aerial imagery within these platforms.

Automated Data Pipeline: Establishing an automated pipeline for data ingestion, preprocessing, model prediction, and output generation to streamline the workflow for end-users.

3. Incorporating Additional Prediction Models:

Species Classification: Extending the current tree crown detection model to include species classification based on the detected crowns, possibly by training on a dataset with labeled tree species or integrating existing botanical databases.

Health Assessment Models: Implementing models to assess the health of detected trees, potentially using additional data sources such as multispectral or hyperspectral imagery.

Integration with Environmental Models: Combining tree crown detection with other environmental

prediction models (e.g., models predicting soil moisture, fire risk, or habitat suitability) to provide more comprehensive ecological assessments.

4. Community and Data Contributions:

Open Data Initiatives: Encouraging the community to contribute annotated datasets to enhance model performance across various forest types and conditions. Model Sharing and Collaboration: Establishing a platform for researchers and practitioners to share custom models and configurations, fostering collaboration and the exchange of best practices.

5. Research and Development:

Exploring New Technologies: Staying updated with advancements in deep learning and computer vision to incorporate cutting-edge techniques into the DeepForest package.

Cross-Domain Applications: Investigating the application of developed models and techniques in related domains, such as agriculture or urban planning, to broaden the project's impact and applicability.

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